

- 6 (a) 92 protons in the nucleus and 92 electrons around nucleus B1
 143 neutrons (in the nucleus) B1 [2]
- (b) (i) α -particle travels short distance in air B1 [1]
- (ii) very small proportion in backwards direction / large angles B1
 majority pass through with no /small deflections B1
 either most of mass is in very small volume (nucleus) and is charged or most of atom is empty space B1 [3]
- (c) $I = Q / t$ C1
 $n/t = (1.5 \times 10^{-12}) / (2 \times 1.6 \times 10^{-19})$ C1
 $n/t = 4.7 \times 10^6 \text{ s}^{-1}$ A1 [3]